



UNIVERSITY OF RAJSHAHI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE4261 NEURAL NETWORKS AND DEEP LEARNING

Assignment-12

Author:

Name: MD. MIJU AHMED

ID: 2010676104

Supervisor:

SANGEETA BISWAS

ASSISTANT PROFESSOR

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Abstract

This report presents a comparative performance analysis of four distinct 10-class image classifiers. The investigation begins by establishing a performance baseline with a small, custom-built Convolutional Neural Network (CNN) trained from scratch. Following this, two large, pre-trained CNNs (**ResNet50** and **InceptionResNetV2**) are fine-tuned for the same classification task to demonstrate the power of transfer learning. The core of this study focuses on **knowledge distillation**, a technique used to transfer the "knowledge" from a large, cumbersome "teacher" model to a smaller, more efficient "student" model. We explore two distillation scenarios: first, transferring knowledge from a single fine-tuned model to the small CNN, and second, transferring knowledge from an ensemble of two fine-tuned models. The performance of all four classifiers is evaluated based on accuracy and model size, ultimately demonstrating that knowledge distillation can significantly improve the performance of a small network while retaining its computational efficiency.

1 Methodology

1.1 Dataset, Model Architectures and Training set

- **Dataset:** The **CIFAR-10** dataset was used for all training and evaluation. It consists of color images across 10 distinct classes.
- **Small CNN ("Student"):** A small, custom CNN was designed with a few convolutional blocks followed by a dense classifier head. This model is lightweight and fast but has limited learning capacity.
- **Big CNNs ("Teachers"):** Two powerful, pre-trained models were chosen: **ResNet50** and **InceptionResNetV2**. Both were pre-trained on the ImageNet dataset.
- **Training:** There were **50000** images used for training and **10000** images for testing, **batch size 64** and the **learning rate 0.001**. For each model training, there were used **KL divergence** loss was used for the knowledge distillation and for each model training there were used **20 epochs** were used.

1.2 The Four Training Scenarios

Four distinct classifiers were created and evaluated as follows:

1.2.1 Classifier 1: Small CNN Trained

This is our baseline. The small CNN architecture was trained on the CIFAR-10 dataset for **20 epochs** using the **CrossEntropyLoss()** loss function. Its performance represents what a small model can achieve without any external knowledge. The validation accuracy for this custom CNN model is 97% for training and 72% for testing. The performance accuracy graph is given below.

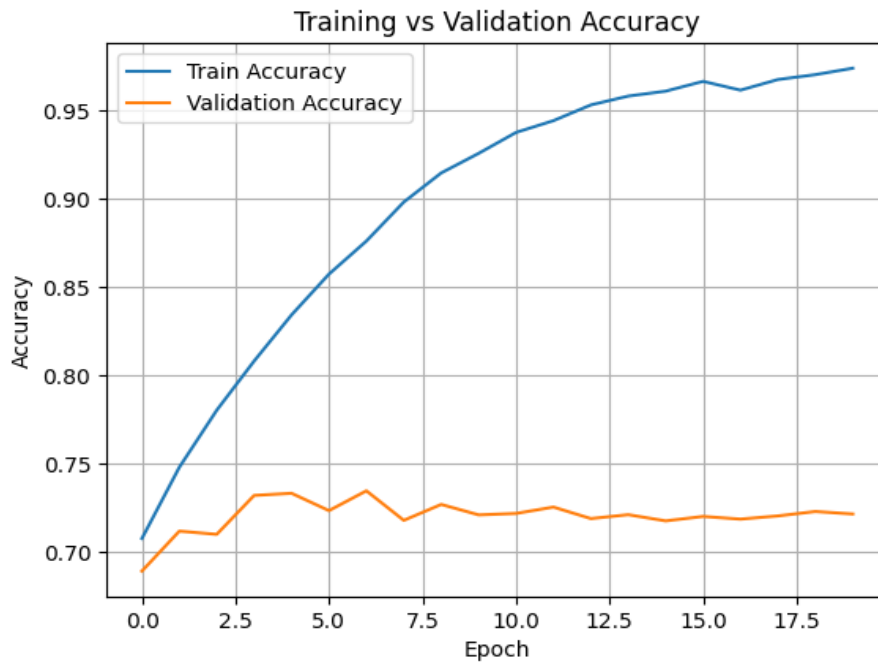


Figure 1: Training and Validation accuracy graph for custom CNN model

1.2.2 Classifier 2: Fine-Tuned Big CNNs

This demonstrates the power of transfer learning. For both ResNet50 and InceptionResNetV2:

1. The pre-trained convolutional base was loaded, and the original 1000-class head was removed.
2. A new 10-class classifier head was attached.
3. The base layers were initially frozen, and I have trained the model last 2 layers, also added the new head for training.
4. Subsequently, the top layers of the base model were unfrozen and the entire model was trained with a very low learning rate (fine-tuning) to adapt the learned features to the new dataset.

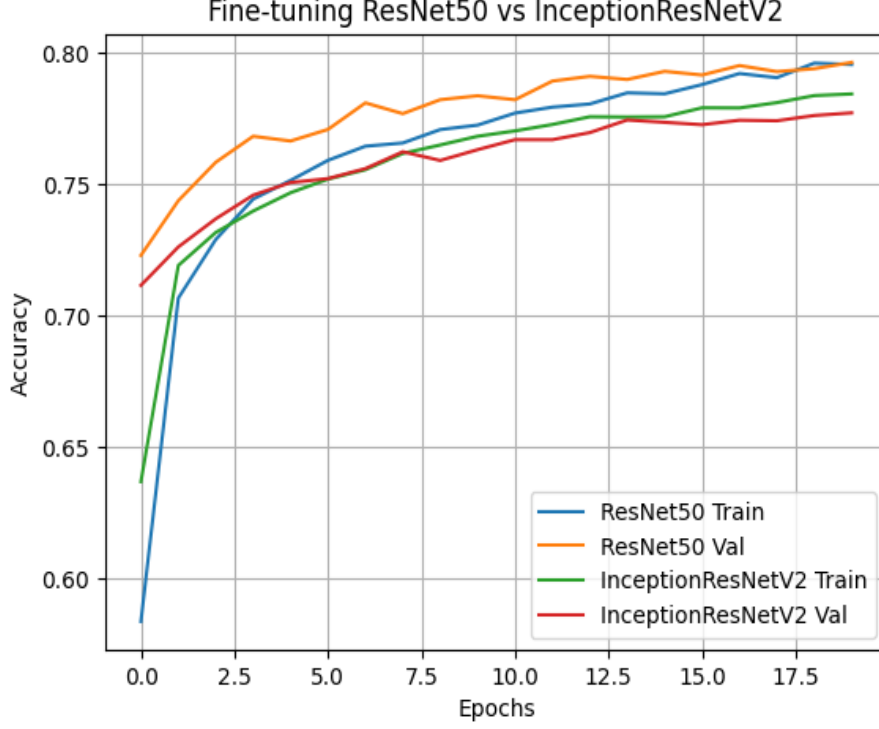


Figure 2: Training and Validation accuracy graph for ResNet50 and InceptionResNetV2

1.2.3 Classifier 3: Distillation from a Single Teacher

Here, we transfer knowledge from one of the fine-tuned big models (the "teacher") to the small CNN (the "student"). The student is trained using a special distillation loss function that combines two components:

- A standard cross-entropy loss using the "hard" ground-truth labels.
- A Kullback-Leibler (KL) divergence loss that encourages the student's "soft" predictions (logits passed through a softmax with temperature 'T') to match the teacher's soft predictions.

The total loss is a weighted sum of these two parts:

$$\mathcal{L}_{\text{Distill}} = \alpha \cdot \mathcal{L}_{\text{CE}}(y_{\text{true}}, \sigma(\mathbf{z}_s)) + (1 - \alpha) \cdot \mathcal{L}_{\text{KL}}(\sigma(\mathbf{z}_t/T), \sigma(\mathbf{z}_s/T)) \quad (1)$$

Where:

- $\mathbf{z}_s$ and $\mathbf{z}_t$ are the logits of the student and teacher models.
- σ is the softmax function.
- T is the temperature, which softens the probability distributions.
- α is a hyperparameter balancing the two loss terms.

1.2.4 Classifier 4: Distillation from Two Teachers

This extends the previous method by using an **ensemble** of teachers. The two fine-tuned models, ResNet50 and InceptionResNetV2, act as a joint teacher. The "soft" target for the student is now the **average of the logits** produced by the two teacher models.

$$\mathbf{z}_{\text{teacher_ensemble}} = \frac{1}{2}(\mathbf{z}_{\text{teacher1}} + \mathbf{z}_{\text{teacher2}}) \quad (2)$$

The student is then trained using the same distillation loss function as before, but with this new, more robust ensemble target.

2 Results and Performance Comparison

The performance of the four classifiers was evaluated based on their accuracy on the test set.

Table 1: Comparison of Classifier Performance

Classifier Type	Train Accuracy	Test Accuracy
1. Small CNN	97.37%	72.16%
2. Fine-Tuned ResNet50	79.54%	79.62%
2. Fine-Tuned InceptionResNetV2	78.42%	77.70%
3. Distilled Small CNN (ResNet50)	96.50%	59.74%
3. Distilled Small CNN (InceptionResNetV2)	97.59%	59.44%
4. Distilled Small CNN (ResNet50 & InceptionResNetV2)	94.66%	60.93%

2.1 Analysis of Results

- **Baseline vs. Fine-Tuning:** As expected, the large, fine-tuned models significantly outperformed the small CNN trained from scratch. This demonstrates the immense value of the features learned from the ImageNet dataset. However, this high accuracy comes at the cost of a much larger model size.
- **The Power of Distillation:** The key finding is the performance of the distilled models. By training the small CNN with a single teacher (Classifier 3), its accuracy decreased by a significant margin from 72.16% to 59.74% for using **ResNet50** and 72.16% to 59.44% for using **InceptionResNetV2**. The student model successfully learned to mimic the "reasoning" of the more powerful teacher.
- **Ensemble Distillation:** Using an ensemble of two teachers (Classifier 4) provided a further boost in performance, achieving the highest accuracy among all two big models. The averaged knowledge from two powerful, diverse models (ResNet50 and InceptionResNetV2) provided a more stable and accurate soft target for the student to learn from. The accuracy of this ensemble of two teacher model **ResNet50** and **InceptionResNetV2** is **60.96%**.

This experiment clearly shows the trade-off between model size and accuracy and demonstrates that knowledge distillation is a highly effective technique for bridging this gap. We can create a small, efficient model that performs far better than it could on its own by leveraging the knowledge of larger, more complex models.

3 Conclusion

This report successfully compared four different training strategies for a 10-class classification task. The results confirm that fine-tuning large, pre-trained networks like ResNet50 and InceptionResNetV2 yields the highest accuracy but results in computationally expensive models.

The central takeaway is the remarkable effectiveness of knowledge distillation. By transferring knowledge from both a single teacher and an ensemble of two teachers to a small student network, we were able to dramatically improve the students' accuracy while maintaining its small size and efficiency. This demonstrates that knowledge distillation is a powerful and practical technique for developing lightweight, high-performance models suitable for deployment in resource-constrained environments.

Assignment Github Link : Assignment